AMOUNT OF AIR BLOWN FROM VENTS DOES NOT CHANGE [MANUAL AIR CONDITIONER]

id0703c2700400

DESCRIPTION	• Malfunction in blower system
POSSIBLE CAUSE	• HEATER 40 A fuse malfunction • Blower unit malfunction • Open or short circuit in wiring harness between fan switch and blower relay • Blower relay malfunction • Blower motor malfunction • Resistor malfunction • Fan switch malfunction

- When performing an asterisked (*) troubleshooting inspection, shake the wiring harness and connectors while performing the inspection to discover whether poor contact points are the cause of any intermittent malfunctions. If there is a problem, inspect make sure connectors, terminals and wiring harnesses are connected correctly and undamaged.

Diagnostic procedure

Step	Inspection		Action
1	**INSPECT FUSE** • Remove the HEATER 40 A fuse. • Is the fuse normal?	Yes	Go to the next step.
		No	**If the fuse is melted:** Repair or replace wiring harness for short to GND and replace the fuse. **If the fuse is deterioration:** Replace the fuse. Then go to Step 16.
2	**INSPECT TO SEE WHETHER MALFUNCTION IS IN BLOWER UNIT OR ELSEWHERE** • Switch the ignition ON (engine off or on). • Turn the fan dial clockwise. • Recirculate air inside the vehicle. • Does the blower motor rotate smoothly?	Yes	Go to the next step.
		No	Go to Step 4.
3	**INSPECT BLOWER UNIT INTAKE VENT** • Is blower unit intake vent clogged?	Yes	Remove obstruction, then go to Step 16.
		No	Inspect if there are any obstruction in the A/C unit passage, then go to Step 16.
4*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN BLOWER RELAY SYSTEM OR RESISTOR SYSTEM** • Switch the ignition ON (engine off or on). • Turn the fan dial clockwise. • Measure the voltage at the following blower motor terminal. — Terminal A (blower motor operation signal) • Is voltage **approx. 12 V**?	Yes	Go to Step 10.
		No	Go to the next step.
5*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN BLOWER MOTOR** • Switch the ignition ON (engine off or on). • Turn the fan dial clockwise. • Measure the voltage at the following blower motor terminal. — Terminal B (blower motor operation signal) • Is voltage **approx. 12 V**?	Yes	Go to the next step.
		No	Go to Step 7.
6*	**INSPECT BLOWER MOTOR** • Inspect the blower motor. (See BLOWER MOTOR INSPECTION.) • Is the blower motor normal?	Yes	Go to Step 16.
		No	Replace the blower motor, then go to Step 16. (See BLOWER MOTOR REMOVAL/INSTALLATION.)

Step	Inspection		Action
7*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN FUSE BLOCK AND BLOWER RELAY) OR ELSEWHERE** • Measure the voltage at the following blower relay terminals. — Terminal A (B+ signal) — Terminal D (B+ signal) • Is the voltage **approx. 12 V**?	Yes	Go to the next step.
		No	**If there is a common connector:If there is no common connector:** Refer to the wiring diagram and verify whether or not there is a common connector between blower relay terminal and HEATER 40 A fuse. • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. • Repair or replace the wiring harness. Go to Step 16.
8*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN BLOWER RELAY AND GROUND) OR ELSEWHERE** • Switch the ignition ON (engine off or on). • Measure the voltage at the following blower relay terminal. — Terminal B (GND signal) • Is the voltage **approx. 0 V**?	Yes	Go to the next step.
		No	**If there is a common connector:If there is no common connector:** Refer to the wiring diagram and verify whether or not there is a common connector between blower relay terminal and body control module (BCM) terminal. • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. • Repair or replace the wiring harness. Go to Step 16.
9*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN BLOWER RELAY AND RESISTOR) OR BLOWER RELAY** • Switch the ignition ON (engine off or on). • Measure the voltage at the following blower relay terminal. — Terminal C (blower motor operation signal) • Is the voltage **approx. 12 V**?	Yes	Repair the wiring harness for lack of continuity between the blower relay and resistor, then go to Step 16.
		No	Replace the blower relay, then go to Step 16.
10*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN BLOWER MOTOR AND RESISTOR) OR ELSEWHERE** • Switch the ignition ON (engine off or on). • Measure the voltage at the following terminal of resistor. — Terminal A (blower motor operation signal) • Is voltage **approx. 12 V**?	Yes	Go to the next step.
		No	**If there is a common connector:If there is no common connector:** Refer to the wiring diagram and verify whether or not there is a common connector between blower motor terminal and resistor terminal. • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. • Repair or replace the wiring harness. Go to Step 16.
11*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN FAN SWITCH AND GROUND) OR ELSEWHERE** • Disconnect the negative battery cable. • Inspect for continuity at the following terminals between the fan switch and ground. — Terminal A (blower motor operation signal) —ground • Is there continuity?	Yes	Go to the next step.
		No	**If there is a common connector:If there is no common connector:** Refer to the wiring diagram and verify whether or not there is a common connector between resistor terminal and ground. • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. • Repair or replace the wiring harness. Go to Step 16.
12	**INSPECT BLOWER UNIT** • Inspect the fan in blower unit. — Is the fan free of interference with the blower unit case? — Is the fan free of foreign material and obstruction? • Is the fan normal?	Yes	Go to the next step.
		No	Remove obstruction, repair or replace the fan and blower unit case, then go to Step 16.

Step	Inspection		Action
13*	**INSPECT RESISTOR** • Inspect the resistor. (See RESISTOR INSPECTION.) • Is the resistor normal?	Yes	Go to the next step.
		No	Replace the resistor, then go to Step 16. (See RESISTOR REMOVAL/INSTALLATION.)
14*	**INSPECT FAN SWITCH** • Inspect the fan switch. (See FAN SWITCH INSPECTION [MANUAL AIR CONDITIONER].) • Is the fan switch normal?	Yes	Go to the next step.
		No	Replace the fan switch, then go to Step 16. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [MANUAL AIR CONDITIONER].)
15*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY BETWEEN RESISTOR AND FAN SWITCH) OR ELSEWHERE** • Switch the ignition off. • Disconnect fan switch connector. • Inspect for continuity at the following terminals between the resistor and fan switch. — Terminal C—C (blower motor speed control signal) — Terminal B—E (blower motor speed control signal) — Terminal D—F (blower motor speed control signal) — Terminal A—D (blower motor speed control signal) • Is there continuity?	Yes	Go to the next step.
		No	**If there is a common connector:If there is no common connector:** Refer to the wiring diagram and verify whether or not there is a common connector between resistor terminal and fan switch terminal. • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. • Repair or replace the wiring harness. Go to the next step.
16	**CONFIRM THAT MALFUNCTION SYMPTOM DOES NOT RECUR AFTER REPAIR** • Is air discharged from vent?	Yes	Troubleshooting completed. Explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if the malfunction recurs.